

Restaurant Analytics Dashboard

Abstract

This project focuses on analyzing restaurant data to identify high-performing restaurants and support data-driven recommendations for a restaurant aggregator platform. Using Tableau for visualization and Excel for data preparation, this study explores key performance indicators such as ratings, average cost, votes, and service availability to identify potential “star” restaurants.

Problem Statement

A restaurant aggregator aims to revamp its B2C portal by leveraging intelligent automation technology. This requires identifying high-performing restaurants based on multiple performance metrics and understanding customer behavior patterns through data analysis.

Project Objectives

- Analyze restaurant performance across multiple cities.
- Identify popular cuisines served across restaurants.
- Understand rating distribution and influencing factors.
- Examine relationship between cost, votes, and ratings.
- Determine online delivery and table booking trends.
- Develop an interactive dashboard to identify star restaurants.

Dataset Description

The dataset includes restaurant-level data such as restaurant name, city, cuisines served, aggregate rating, number of votes, average cost for two, and service availability (online delivery and table booking).

Methodology

Data cleaning was performed to remove duplicates and handle missing values. The cuisines column containing multiple values was normalized by splitting individual cuisines into separate rows to enable accurate analysis. Exploratory Data Analysis (EDA) was conducted to identify trends and patterns.

Exploratory Data Analysis

- Geographical distribution of restaurants across cities.
- Top 10 cuisines served across locations.
- Rating distribution analysis.
- Online delivery and table booking availability.
- Cost vs Rating relationship analysis.
- Votes vs Rating relationship analysis.

Dashboard Features

- KPI Metrics: Average Rating, Average Cost, Total Restaurants, Online Delivery Percentage.
- Interactive Filters: City, Cuisine, Price Range, Delivery Status, Table Booking.
- Map Visualization for restaurant distribution.
- Scatter plots for relationship analysis.
- Bar chart for cuisine popularity.
- Donut chart for delivery service availability.

Key Insights

- Higher customer engagement (votes) tends to correlate with higher ratings.
- Moderate-cost restaurants generally maintain better ratings.
- Online delivery services improve restaurant accessibility.
- Popular cuisines dominate restaurant presence across cities.

Business Recommendations

- Promote restaurants with high ratings and customer engagement.
- Encourage adoption of online delivery services.
- Focus on high-demand cuisines for expansion strategies.
- Use rating and vote metrics for recommendation algorithms.

Tools & Technologies Used

- Tableau – Dashboard Development
- Microsoft Excel – Data Cleaning
- Python – Data Preparation

Conclusion

The developed dashboard provides valuable insights into restaurant performance and enables stakeholders to make informed business decisions. It supports the identification of star restaurants and enhances recommendation systems through data-driven analytics.